Exercise 2: Military coups

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

(a) We read the data set from coups.txt file and is stored in the variable data. Convert political liberalization levels into a factor, we treat it as a categorical variable. We also inspect the variable data

```
## Rows: 36
## Columns: 9
## $ miltcoup  <int> 5, 6, 2, 0, 1, 3, 1, 2, 3, 0, 0, 1, 1, 1, 5, 1, 1, 0, 1, 1, ~
## $ oligarchy <int> 7, 13, 13, 0, 0, 14, 15, 0, 2, 0, 0, 8, 10, 0, 17, 6, 0, 0, ~
## $ pollib    <fct> 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 1, 0, ~
## $ parties   <int> 34, 62, 10, 34, 5, 14, 27, 4, 27, 37, 2, 21, 17, 15, 22, 19,~
## $ pctvote   <dbl> 45.68, 17.50, 34.39, 30.30, 30.46, 16.20, 44.16, 52.00, 40.8~
## $ popn      <dbl> 4.600, 8.800, 5.300, 11.600, 0.361, 3.000, 5.500, 0.458, 2.2~
## $ size      <dbl> 113, 274, 28, 475, 4, 623, 1284, 2, 342, 322, 23, 28, 1222, ~
## $ numelec   <int> 8, 5, 3, 14, 2, 6, 4, 6, 10, 12, 4, 4, 6, 12, 6, 9, 0, 6, 1,~
## $ numregim  <int> 3, 3, 3, 3, 1, 4, 3, 2, 4, 2, 2, 4, 3, 2, 3, 3, 1, 3, 4, 3, ~
```

We perform a Poisson regression analysis on the dataset and display a summary of the model

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib + parties + pctvote +
##     popn + size + numelec + numregim, family = poisson, data = data)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.2334274  0.9976112  -0.234  0.81500
## oligarchy    0.0725658  0.0353457   2.053  0.04007 *
## pollib1     -1.1032439  0.6558114  -1.682  0.09252 .
## pollib2     -1.6903057  0.6766503  -2.498  0.01249 *
## parties      0.0312212  0.0111663   2.796  0.00517 **
## pctvote      0.0154413  0.0101027   1.528  0.12641
## popn         0.0109586  0.0071490   1.533  0.12531
## size        -0.0002651  0.0002690  -0.985  0.32444
## numelec     -0.0296185  0.0696248  -0.425  0.67054
## numregim     0.2109432  0.2339330   0.902  0.36720
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
```

```
##
##     Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 28.249  on 26  degrees of freedom
## AIC: 113.06
##
## Number of Fisher Scoring iterations: 5
```

The Poisson regression model shows that oligarchy, pollib2, and parties are significant predictors of military coups ($p < 0.05$). Specifically, more years of military oligarchy and more legal political parties increase coups, while higher political liberalization (pollib2) reduces coups. Non-significant predictors (pctvote, popn, size, etc.) can be considered for removal to simplify the model.

(b) We start reducing the explanatory variables in the model by using the step down approach

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib + parties + pctvote +
##     popn + size + numregim, family = poisson(link = "log"), data = data)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.4577458  0.8602345  -0.532  0.59464
## oligarchy    0.0812015  0.0288154   2.818  0.00483 **
## pollib1     -0.9642976  0.5620939  -1.716  0.08625 .
## pollib2     -1.5149509  0.5269441  -2.875  0.00404 **
## parties      0.0293409  0.0103101   2.846  0.00443 **
## pctvote      0.0139115  0.0094654   1.470  0.14164
## popn         0.0099592  0.0067249   1.481  0.13862
## size        -0.0002688  0.0002687  -1.000  0.31710
## numregim     0.1804415  0.2241166   0.805  0.42075
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 28.430  on 27  degrees of freedom
## AIC: 111.24
##
## Number of Fisher Scoring iterations: 5
```

Number of legislative and presidential elections has a p-value of 0.670 so we remove it from the model and re-evaluate the mode we keep on doing this till we can no longer eliminate any other variable Iteratively we remove the explanatory variables (in order) number of regime types, area in 1000 square km, population in millions in 1989.

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib + parties + pctvote,
##     family = poisson(link = "log"), data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.116499   0.513751  -0.227  0.82061
## oligarchy    0.094712   0.023184   4.085  4.4e-05 ***
## pollib1     -0.620756   0.487526  -1.273  0.20292
```

```
## pollib2     -1.310374   0.489017  -2.680  0.00737 **
## parties      0.025745   0.009552   2.695  0.00704 **
## pctvote      0.012057   0.009072   1.329  0.18383
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 31.069  on 30  degrees of freedom
## AIC: 107.88
##
## Number of Fisher Scoring iterations: 5
```

The next variable to remove is a categorical variable.We recode the pollib variable to combine levels 0 and 1 into a single group (0) and treat level 2 as a separate group (1). This simplifies the model by focusing on the significant difference between pollib = 2 (higher political liberalization) and the combined group of pollib = 0 and 1 (lower liberalization), while reducing complexity and improving interpretability.

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib_recoded + parties +
##     pctvote, family = poisson(link = "log"), data = data)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.496647   0.445284  -1.115  0.26470
## oligarchy       0.088765   0.021957   4.043 5.28e-05 ***
## pollib_recoded -0.789590   0.289344  -2.729  0.00635 **
## parties         0.025184   0.009391   2.682  0.00733 **
## pctvote         0.009914   0.008713   1.138  0.25519
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 32.542  on 31  degrees of freedom
## AIC: 107.35
##
## Number of Fisher Scoring iterations: 5
```

The we remove the variable percent voting in last election since it is insignificant p-value= 0.255

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib_recoded + parties,
##     family = poisson(link = "log"), data = data)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.153867   0.307817  -0.500   0.6172
## oligarchy       0.086951   0.021593   4.027 5.65e-05 ***
## pollib_recoded -0.717419   0.285632  -2.512   0.0120 *
## parties         0.022562   0.009038   2.496   0.0125 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 33.818  on 32  degrees of freedom
## AIC: 106.63
##
## Number of Fisher Scoring iterations: 5
```

The final model simplifies the initial model by retaining only significant predictors (oligarchy, pollib_recoded, parties). Both models show that oligarchy and parties increase coups, while pollib_recoded (or pollib2 in the initial model) reduces coups.The final model has a lower AIC (106.63 vs. 113.06), indicating a better balance of fit and simplicity. Removing non-significant predictors improves interpretability without sacrificing key insights.

(c)We find the expected number of Coups for different Political Liberalization levels we consider pollibs 0 and 1 to not have any significant difference

```
## Average oligarchy: 5.222222
```

```
## Average parties: 17.08333
```

```
## Predicted number of coups for None or Limited Civil Rights): 1.985048
```

```
## Predicted number of coups for Full Civil Rights: 0.9687238
```

Countries with full civil rights (pollib_recoded = 1) are predicted to have fewer military coups (0.969) compared to those with none or limited civil rights (pollib_recoded = 0, 1.985 coups). oligarchy and parties increase coups, while pollib_recoded significantly reduces them.